With receipts: twelve checked cases from agent-assisted audits of palaeobiology's data and software

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Abstract

Quantitative palaeobiology rests on shared infrastructure: occurrence databases, published data deposits, time scales, plate models and analysis packages. Problems in these layers can pass into published inference unnoticed, because auditing them is slow, broad and rarely rewarded. During the first month of a research programme run by two AI systems (Anthropic's Claude and OpenAI's Codex) with a human facilitator, agent-assisted audits turned up candidate findings across this infrastructure. We report twelve of them as a case series. Each was chosen for its consequence for inference, replayed from pinned inputs, and checked by a different actor along a separate computational route. The cases cover:

- deposited occurrences absent from a dated public database snapshot (71,103 of 368,760 molluscan records);
- interval ages that change between archived database snapshots;
- excess rows in a database update that changed 1,388 of 2,267 exposure means in one of our own analyses;
- stored sample ages that disagree with their own tie points;
- three implementation defects in analysis packages;
- a calibration failure, alongside two positive controls;
- the semantics of a widely used extinction-time interval;
- model-dependent palaeolatitudes;
- two published spreadsheets that differ from their sources;
- a printed significance value that its stated test does not reproduce (0.316, not 3.4×10^{-16});
- archived calibration scores that do not replay.

Every proposition is bounded to the version, design and convention tested. Several original headlines were narrowed or withdrawn during replay, and we report those corrections. The series does not estimate how common such problems are, and it does not show that agent-assisted auditing is better than other auditing. It does show, with receipts anyone can rerun, the kinds of problem that cheap, systematic replay exposes, and what each one does and does not change for inference.

1. Introduction

A palaeobiological result is usually several layers of other people's work deep. An occurrence count depends on a database query and that database's state on the day of the query. An age depends on a time scale and an age model. A palaeolatitude depends on a plate model and its reference frame. A rate depends on an estimator and on the package that implements it. Each layer is maintained by people with too little time. Each changes, and few users check the layer below the one they work in.

The community knows this. In the recent survey of open questions in palaeontology (Smith et al. 2025), several of the priorities concern exactly this layer: quantifying collecting and preservation biases, finding and correcting data-entry biases, accounting for data loss, and making data open and combinable. Audits of the infrastructure answer those calls directly. They are rarely done, because they are tedious, broad and thankless, and the person best placed to find a problem is usually the person with the least time to look for it.

Agentic AI systems change the cost of that work. During October and November 2026, the Extinction and Palaeontology Research Programme (EPRP) used two such systems, with a human facilitator, to vet research questions against the data that could answer them. Vetting meant mirroring databases and deposits, benchmarking packages against known truth, and replaying published numbers. That work produced 189 candidate findings about the infrastructure itself.

This paper reports twelve of them as a case series. For each case we state the checked proposition, the evidence and how it was replayed, what it changes for inference, and what it does not show. We also report what the auditing agents themselves got wrong. The aim is not a rate or a ranking. It is a set of inspectable cases, each with a command that reproduces it.

2. How the cases were found, chosen and checked

Finding. The candidates came from read-only mirrors of databases, deposits and plate models (each hashed on arrival), benchmark suites declared before they were run, and checks on published numbers made while vetting research questions. Three working sessions, one each for databases, packages and references, recorded each finding with its evidence location and the checks it had received.

Choosing. From the 189 candidates we chose twelve cases by consequence for inference: what each would change in an analysis someone plausibly runs. Being a defect was not enough on its own. We also wanted the selection to cover the kinds of finding in the pool, and we included positive controls, because a series that only finds faults says nothing about how discriminating the checks were. Licence and bibliographic-metadata problems were left out as having little bearing on inference. Location data whose description could itself cause harm were left out entirely. The selection note, with reserves, is in the repository (Supplement S1).

Checking. Each case was replayed from pinned inputs on 8 November 2026. A different actor from the one who made the original finding then rederived each supporting proposition by a separate computational route: a separate implementation, a recount from raw files, or a closed-form calculation. Each proposition is a row in one claims ledger, with its value, evidence files, the commit at which the evidence was verified, a command that regenerates it, and a statement of what it does not show. A row is marked supported only after that independent check. Where the replay did not reproduce an original number, the number was withdrawn, and the withdrawal is recorded (Section 4; Supplement S3).

Classes. Following the audit design, each case is classified by what kind of finding it is:

- an implementation defect;
- a data inconsistency;
- a calibration failure under a stated model;
- a semantic finding (what a quantity actually estimates);
- an intended version or model choice that users need to know about;
- a replay failure;
- a sensitive access finding.

These classes matter. An intended version change is not an error, and calling it one would be unfair to the people who maintain the resource.

3. The cases

Table 1. The twelve cases. Values are from the claims ledger; each row's conditions are in the text.

#	Case	Class	Checked proposition	Claims
1	Deposited records absent from a dated public snapshot	sensitive access finding	71,103 of 368,760 molluscan occurrences in a published deposit are not in a public snapshot of 23 October 2026	C191, C192
2	Interval ages change between archived snapshots	intended version choice	460, 317 and 17 intervals changed modal base age across three snapshot brackets	C076
3	Excess rows in a database update	data inconsistency	28,832 excess rows under a specified key; removing 16,260 changed 1,388 of 2,267 exposure means	C043, C045
4	Stored ages disagree with their tie points	data inconsistency	4,111 of 61,862 samples differ by more than 0.01 Myr	C040
5	Age–depth conversion above hiatuses	implementation defect	3,972 of 8,282 test points wrong, by up to 0.466 Myr	C096

#	Case	Class	Checked proposition	Claims
6	Numbers compared as text	implementation defect	at a threshold of 5, differences of 10 to 499 were not flagged	C102
7	Preservation-rate calibration, with positive controls	calibration failure	intervals covered the truth in 81 of 100 and 33 of 50 replicates; classical bounds near nominal	C107, C194, C196, C105, C111
8	What an extinction-time interval estimates	semantic finding and defect	the interval reflects resampled dating error only; the call overwrites the caller's random-number state	C110, C098
9	Plate-model palaeolatitudes	intended model choice	median spread among three models from 2.9 degrees at 10 Ma to 13.6 degrees at 400 Ma	C125, C123
10	Published spreadsheets differ from their sources	data inconsistency	2,967 of 6,562 capture histories altered; 220 rows shifted one column	C035, C048
11	A printed probability	arithmetic error in a published claim	the stated binomial test gives 0.316, not 3.4×10^{-16}	C190
12	Archived calibration scores	replay failure	three saved scores differ from rescored values by 42.4, 17.5 and 11.9	C195, C193

3.1 Data access and provenance

Case 1. A published global analysis of marine latitudinal diversity (Benson et al. 2025) deposited its occurrence data. Of the 368,760 occurrences in the deposit's molluscan member, 71,103 are not present in a public Paleobiology Database snapshot taken on 23 October 2026. That is about one record in five. Absence from a dated snapshot does not by itself establish why the records are absent; deletion, merging and other database changes can all contribute. It does mean that a reader who rebuilds the analysis from today's public database cannot obtain the same input. The consequence showed up in our own work. A cheap screen for outside-Europe and North America tropical support passed in 14 of 21 stages with the full deposit, and in 12 with snapshot members only. We report these records as counts only.

Case 2. Archived query products from the Paleobiology Database show interval base ages changing between consecutive snapshots. Across three main brackets, 460, 317 and 17 intervals with at least 50 single-interval occurrences changed their modal base age. If the 50-occurrence threshold must hold in both snapshots, the middle count is 316. A query repeated on different dates can therefore return different ages for the same interval. We have not shown whether the database keeps historical versions accessible, and we make no comparison here with the international chronostratigraphic chart. This is a version change, and the point for users is to record the date and version of every query.

Case 3. The 2023 update of the Triton planktonic-foraminifera database (Fenton et al. 2021) contains 28,832 excess rows under the key we specified. A conservative recipe that keeps one row per key by source priority removes 16,260 of them. Rows that share a key are not necessarily repeated physical observations, so we call them excess rows under that key, not duplicates. The consequence is measured in one of our own analyses. Rebuilding EPRP's exposure index with and without the removal changed 1,388 of 2,267 comparable lineage-bin means, 411 of them by at least one degree. Of 491 lineage-pair rows, 8, 8, 2 and 1 changed their classification under four different classification rules. An earlier internal report of 1,454 and 423 changed means did not reproduce under the pinned recipe and is withdrawn (Supplement S3).

Case 4. The final snapshot of the Neptune Sandbox Berlin database (Renaudie, Lazarus, and Diver 2020) stores an age for each sample. We recomputed each age from its hole's current tie points by linear interpolation. Of 61,862 samples, 4,111 in 56 holes differ from the stored value by more than 0.01 Myr. The recomputation takes the first age where two tie points share a depth; with the other choice, 8 of the differences disappear. Neither set of ages is thereby shown to be right. A user who takes the stored ages and one who recomputes from the tie points will be working from different numbers.

3.2 Software correctness

Case 5. admtools 0.7.0 converts stratigraphic height to time. On a synthetic age–depth model with two hiatuses, 3,972 of 8,282 random query points received wrong times, by up to 0.466 Myr, against exact linear-segment truth; a targeted probe gave an error of about 2.7 Myr. The errors fall in affected interpolation segments. Later segments can be correct, so the error does not apply to every point above a hiatus.

Case 6. fossilbrush 1.0.6 flags occurrences whose ages differ from a reference range by more than a threshold. It compares the differences with the threshold as text. At a threshold of 5, it failed to flag differences of 10, 20, 49.9, 100 and 499. A character comparison predicts every flag on the test grid, which identifies the mechanism. This is a tested mechanism on a deterministic grid, not a measured rate of misflagged records in real data.

Case 8. The GRIWM estimator of extinction time (Bradshaw et al. 2012), as implemented in Rextinct 1.1.0, reports an interval whose width comes only from resampling the input dating errors. With zero dating error, the interval collapses to a point. It is therefore not an interval for the extinction time given the sampling of the record, and it should not be reported as one. The same implementation also overwrites the calling program's random-number state. Any analysis that draws random numbers after calling it inherits a stream it did not choose. We have not quantified the effect of either finding on any published study.

3.3 Calibration, with positive controls

Case 7. We simulated fossil records under a declared design and asked whether PyRate's (Silvestro, Salamin, and Schnitzler 2014) 95% intervals for the preservation rate covered the true value. They covered it in 81 of 100 replicates in one arm and 33 of 50 in a lower-preservation arm, and the posterior medians sat above the truth (median relative bias 0.0895 and 0.458). The intervals are not too narrow: every miss lies below the lower endpoint, which points to location bias rather than understated spread.

The same benchmark showed that the checks can tell good from bad. In the ideal arm, the classical Strauss–Sadler range-end bound (Strauss and Sadler 1989) covered the truth in 1,895 and 1,899 of 2,000 cases in two independent implementations, and the Robson–Whitlock bound in 1,909. The simulation generators matched their analytic moments. Two qualifications travel with these controls:

- **A declared gate failed.** The pre-declared overall agreement gate required 99% of paired bounds to agree within 0.02 Myr; 0.91 did. We report the failure.
- **The hiatus arm deviated from its declaration.** Its executed windows were shifted one Myr from the declared design. In that arm, Strauss–Sadler coverage fell to 1,125 of 1,272 retained cases, with 728 cases dropped.

3.4 Spatial models

Case 9. Three widely used plate models (Merdith et al. 2021; Matthews et al. 2016) place the same 300 public points at different palaeolatitudes. The median spread among them is 2.9 degrees at 10 Ma, 5.7 at 50 Ma and 13.6 at 400 Ma. This is structural disagreement among models, not calibrated error. A diversity gradient analysed in latitude bands of 20 degrees will move points between bands depending on the model chosen. Separately, two routes to reconstructing the same named model differ by an approximately rigid, age-specific rotation. That is consistent with their using different absolute reference frames. We do not identify either frame as correct. The practical point is that a model name alone does not fix the coordinates.

3.5 Published deposits and published numbers

Case 10. The electronic supplement of a study of extinction selectivity (Monarrez, Heim, and Payne 2023) provides capture histories in a spreadsheet as well as an intact binary file. Comparing the two, we found 2,967 of 6,562 histories in the spreadsheet with altered values. 3,595 more can be restored exactly by padding to the known width of 86 characters. The pattern is consistent with spreadsheet software converting long digit strings to numbers, though we did not establish how the file was produced. Separately, the one-file All-Life table of the Fossil Record 2 compilation has 220 rows shifted one column left relative to its chapter

workbooks. In 112 rows this moves a family's range by one interval. We did not rerun either study's analysis, and we make no claim about its conclusions.

Case 11. A recent analysis of cyclicity in stratigraphic boundaries (Rampino and Caldeira 2025) states a binomial test with n = 47, k = 28 and p = 0.55, and prints a tail probability of 3.4×10^{-16}. The stated test gives 0.316. We confirmed this by exact rational arithmetic, by integer polynomial convolution, by a direct exact sum and in R. This does not bear on the paper's separate spectral analysis, which we did not rerun.

Case 12. The INDITEK-2 model of Phanerozoic marine diversity (Cermeño et al. 2022) archives its calibration runs. We reran three archived proposals from the pinned archive. Their predictions replay exactly, to 2,978 cells. Their saved residual sums of squares, however, differ from the rescored values by 42.4, 17.5 and 11.9, and a separate, model-free rescoring agrees with ours. All three are rejected proposals, not accepted draws. The cause of the disagreement is unresolved, and we make no claim about the published conclusions.

4. What the auditing agents got wrong

The agents' own errors are part of the record. Replay withdrew or narrowed several of the original findings:

- **Case 3.** The changed-mean counts of 1,454 and 423 did not reproduce under the pinned recipe, and an 11–13% sample inflation figure was withdrawn.
- **Case 4.** A sample count that included a header row was corrected to 61,862.
- **Case 5.** "Wrong above every hiatus" was false; the error is confined to affected segments.
- **Case 7.** "Too narrow" intervals were not supported.
- **Case 10.** A count of 5,291 cells in scientific notation had been read as a count of damaged histories.

Earlier in the programme, the two agents corrected each other's scientific reasoning repeatedly. One agent's proposed rule, for example, was refuted by the other's counterexample, and a predicted outcome of a screening test was wrong. The full correspondence record is available. These errors were caught because each finding was rechecked by a different actor along a different route. We do not claim that this arrangement catches more errors than any other; the series contains no comparison that could show it.

5. Recommendations

- **For database providers:** expose versions and change logs for interval ages and records; say in query results when records have been removed or merged.
- **For package authors:** test against known truth, including hiatuses and boundary cases; compare numbers as numbers; never reset the caller's random-number state.
- **For authors of deposits:** deposit identifiers and histories as text in machine-readable formats, not in spreadsheets; archive the scorer and observations with any saved calibration scores.
- **For users:** record the date and version of every query; rerun spatial analyses under more than one plate model and state the reference frame; replay a published number before building on it.
- **For programmes that use AI agents:** declare tests before running them, keep failed gates, have a different actor recheck each finding by a different route, and publish the correspondence.

6. Discussion

6.1 What follows

Each case is small on its own. Together they show that the layer beneath published palaeobiology holds problems of several distinct kinds: version changes users need to know about, inconsistencies within a resource, implementation defects, estimators whose output is easy to misdescribe, and published numbers that do not survive replay. None of these problems needed new data or new theory to find. They needed someone to take the time to replay each layer against known truth or against its own source.

6.2 Which conclusions depend on which assumptions

Table 2. Dependence of the main conclusions on their assumptions.

Conclusion	Rests on	Fails if
Each case's proposition holds for the version tested	Pinned inputs and the stated convention	A different input version, convention or design is substituted
The consequence for inference is real	The downstream calculation named in the case	The downstream use differs from the one tested
The independent check adds information	A different actor and computational route	The routes share an error (e.g. a common input misread)
Third-party findings are not already known	Our search of changelogs and issue trackers (incomplete)	A provider documented the behaviour before our audit

6.3 What would sharpen the answer

- **Disclosure responses.** Providers' and authors' replies would establish which findings are already known, intended or fixed.
- **A controlled comparison.** The same audit run by experienced human auditors would allow the cost and yield of agent-assisted auditing to be compared, which this series cannot do.
- **Downstream replays.** Rerunning one or more published analyses that use the affected layers would measure consequence directly, rather than by the proxy analyses used here.

7. Conclusions

Cheap, systematic replay of palaeobiology's shared infrastructure exposes problems of several kinds. The twelve cases here are each bounded, reproducible and checked by a second route. They are a sample of what one programme found, not a measure of how often such problems occur. Their value is that anyone can rerun them, and that providers, authors and users can act on them.

Methods in brief

Supplement S1 gives the case selection and its rules. S2 gives the replay and independent check for each case, with commands. S3 is the audit trail of withdrawn and narrowed statements.

Roles of AI systems and humans

- **Codex (OpenAI GPT-6, with GPT-6.1 Sol delegates)**
  - took over the programme's database, reference and package work from 28 October;
  - replayed all twelve cases;
  - built the claims ledger;
  - made the independent checks.
- **Claude (Anthropic)**, Opus 5.5 in the main session and Claude Sonnet subagents:
  - ran the database, package and reference sessions that produced most of the original findings, before 28 October;
  - compiled the 189-row candidate ledger;
  - chose the twelve cases;
  - wrote the interpretation specification and drafted this paper.
- **Barry W. Brook** facilitated the programme, approved the paper on 25 October 2026, chose this case series for development on 8 November 2026, and holds decisions on provider contact, submission and release.

The two AI systems reviewed each other's work throughout. Their correspondence is published in the repository as reviewed snapshots (`docs/correspondence/`).

Data and code

The repository is public at https://github.com/BWBrook/eprp; code is MIT-licensed and text, figures and data are CC BY-SA 4.0. The claims ledger (`papers/infrastructure-audit/manuscript/claims.csv`) gives, for each proposition, the evidence files, the commit at which they were verified and the command that regenerates them. Third-party sources keep their own licences and are not redistributed. Records absent from public database snapshots are reported as counts only.

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